

Airline Passenger Satisfaction Predictive Modeling & Analysis

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1. Introduction & Problem Definition

The airline industry is one of the most competitive service sectors in the world. Customer satisfaction directly influences brand loyalty, net promoter scores, and long-term revenue retention. Airlines collect large volumes of survey data from millions of passengers yet translating that raw data into actionable predictions remains a challenge.

This project addresses a core classification problem: given a structured passenger survey response, can a machine learning model accurately predict whether a passenger is satisfied or dissatisfied with their overall travel experience?

1.1 Problem Statement

Objective: Build a supervised binary classification pipeline to predict airline passenger satisfaction (satisfied vs. neutral or dissatisfied) from survey responses and flight metadata.

The problem is practically significant because:

- Airlines that can predict dissatisfied passengers early may proactively address complaints, improving retention.
- Identifying the service dimensions most correlated with satisfaction allows targeted operational improvements.
- A deployed prediction interface (Streamlit web app) enables airline staff to instantly evaluate new passenger profiles without re-running the full pipeline.

1.2 Goals & Objectives

- Profile and clean raw airline survey data, ensuring structural integrity before modeling.
- Conduct comprehensive Exploratory Data Analysis (EDA) to surface actionable patterns.
- Engineer a clean numeric feature matrix using label encoding and standardization.
- Train and compare multiple model families: Artificial Neural Networks (ANN), Random Forest, and XGBoost.
- Select the best-performing model based on held-out test accuracy and F1 score.
- Export trained artifacts for deployment in a Streamlit prediction interface.

2. Dataset Description & Justification

The dataset used in this project is a publicly available airline passenger satisfaction survey dataset, commonly referenced in machine learning benchmarks. It contains structured records for over 100,000 passengers, each described by demographic attributes, flight characteristics, and 14 service quality ratings.

2.1 Dataset Overview

Attribute	Details
Source	Airline Passenger Satisfaction Survey (Kaggle)
Target Variable	satisfaction (binary: satisfied / neutral or dissatisfied)
Total Features	22 features after preprocessing
Feature Types	Demographic, operational, and ordinal service ratings (0–5 scale)
Missing Values	Sparse; resolved by listwise deletion (< 0.3% of rows)
Duplicates	Zero confirmed after deduplication check

2.2 Feature Groups

Features fall into three natural groups:

Demographic & Travel Metadata

- Gender, Age, Customer Type (Loyal / Disloyal)
- Type of Travel (Business / Personal), Class (Business / Eco / Eco Plus)
- Flight Distance, Departure Delay in Minutes, Arrival Delay in Minutes

Service Quality Ratings (0–5 ordinal scale)

- Inflight Wi-Fi Service, Ease of Online Booking, Online Boarding
- Food and Drink, Seat Comfort, Inflight Entertainment
- On-board Service, Leg Room Service, Baggage Handling
- Check-in Service, Inflight Service, Cleanliness
- Gate Location, Departure/Arrival Time Convenience

2.3 Justification

This dataset is well-suited to the project goals because:

- It is large enough to support deep learning models without overfitting on training noise.
- Its mix of continuous, ordinal, and categorical features exercises a full preprocessing pipeline.
- The binary target is well-defined and practically meaningful in the airline domain.
- Class balance is not extreme, making accuracy a reliable primary metric alongside F1.

3. Methodology

The project follows a structured end-to-end machine learning workflow composed of five sequential stages: data quality assurance, exploratory data analysis, feature engineering, model training, and evaluation.

3.1 Data Preprocessing

3.1.1 Data Quality Checks

Before any modeling, structural integrity was verified across three dimensions:

Check	Finding	Resolution
Missing Values	Sparse scatter across Arrival Delay column	Listwise deletion (< 0.3% rows affected)
Duplicate Rows	Zero duplicate passenger records	No action required
Dtype Consistency	Service ratings loaded as int64	Cast to object for EDA; float32 post-encoding

3.1.2 Outlier Removal

Outliers in numeric features (Age, Flight Distance, Departure Delay, Arrival Delay) were detected and removed using the Interquartile Range (IQR) method:

|

$$\text{Lower bound} = Q1 - 1.5 \times IQR \quad | \quad \text{Upper bound} = Q3 + 1.5 \times IQR$$

Rows with any numeric value outside these bounds were excluded, reducing the risk of extreme values distorting model gradients or tree splits.

3.2 Feature Engineering & Encoding

All features were converted to numeric form before training. The encoding strategy for each feature type was:

Feature	Encoding Strategy	Rationale
satisfaction (target)	Binary map: 0 = dissatisfied, 1 = satisfied	Required for binary cross-entropy loss
Gender	Binary map: Male=0, Female=1	Simple ordinal-equivalent mapping
Customer Type	sklearn LabelEncoder (saved as dict)	Aligns with Streamlit inference-time mapping
Type of Travel	sklearn LabelEncoder (saved as dict)	Aligns with Streamlit inference-time mapping
Class	sklearn LabelEncoder (saved as dict)	Aligns with Streamlit inference-time mapping
Service ratings (0–5)	Cast to float32	Already numeric ordinal; no further encoding needed

3.3 Train / Validation / Test Split

A nested stratified split was applied to produce three disjoint partitions:

- Test set (20%): held out entirely for final unbiased evaluation.
- Validation set (10% of training portion): used for early stopping and hyperparameter monitoring in neural networks.
- Training set (remaining 70): used for model fitting.

All splits use `random_state=42` for full reproducibility.

3.4 Normalization

A `StandardScaler` was fit exclusively on the training set and applied identically to validation and test sets. This prevents data leakage — scale parameters derived from held-out data must never influence model training or its evaluation.

4. System Design & Interface Explanation

The system is composed of two tightly coupled components: the offline training pipeline (Jupyter Notebook) and the online inference interface (Streamlit web application).

4.1 Training Pipeline Architecture

The notebook implements a linear, reproducible pipeline with seven stages:

Stage	Component	Output
1	Data Loading & Column Pruning	Cleaned DataFrame (id, Unnamed, is_train dropped)
2	Quality Checks (missing, duplicates)	Fully observed design matrix
3	Exploratory Data Analysis (EDA)	Visualizations + correlation heatmap
4	Outlier Removal (IQR)	Outlier-free feature matrix
5	Feature Encoding	Numeric design matrix (float32)
6	Train / Val / Test Split + Scaling	Three reproducible data partitions
7	Model Training & Evaluation	Trained models + classification reports
8	Artifact Export	Pickle files for Streamlit deployment

4.2 Exported Artifacts

Five artifacts are serialized to the artifacts/ directory after training:

File	Contents	Used In
best_model.pkl	Winning trained estimator (ANN / RF / XGBoost)	Streamlit prediction
input_scaler.pkl	Fitted StandardScaler	Feature normalization at inference
class_encoder.pkl	Class label → integer mapping dict	Categorical encoding at inference
customer_encoder.pkl	Customer Type label → integer mapping dict	Categorical encoding at inference
travel_type_encoder.pkl	Travel Type label → integer mapping dict	Categorical encoding at inference

4.3 Web Interface

The deployed application provides an interactive prediction interface accessible from any browser. Its workflow is:

- User inputs passenger details via sidebar controls (dropdowns, sliders, number inputs).
- The app applies the same label-encoding dictionaries and StandardScaler used during training.
- The encoded feature vector is passed to the loaded best_model.pkl.
- The predicted class (Satisfied / Not Satisfied) and confidence probability are displayed in real time.

No retraining occurs at inference time — the app is a pure prediction serving layer.



Figure 1 — Prediction Interface

Plane AIRLINES

Passenger & Trip

SELECT: Male Female

CUSTOMER TYPE: Level Business

TYPE OF TRAVEL: Business Personal

CLASS: Business Eco Plus Eco

AGE: 30 1000

FLIGHT DISTANCE (KM): 0 1000

DEPARTURE DELAY (MIN): 0 1000

ARRIVAL DELAY (MIN): 0 1000

Service Ratings

- Delights WiFi: ★☆☆☆☆
- Departure/Arrival Time: ★☆☆☆☆
- Online Booking: ★☆☆☆☆
- Seat Location: ★☆☆☆☆
- Food & Drink: ★☆☆☆☆
- Online Boarding: ★☆☆☆☆
- Seat Comfort: ★☆☆☆☆
- In-flight Entertainment: ★☆☆☆☆
- On-board Service: ★☆☆☆☆
- Lap Board: ★☆☆☆☆
- Baggage Handling: ★☆☆☆☆

Predict Satisfaction

Neutral / Dissatisfied 0.7%

Figure 2 — Input Panel)

5. Exploratory Data Analysis (EDA)

EDA was structured around four themes: class balance, demographic and travel segmentation, service quality dimensions, and numeric correlations.

5.1 Class Distribution

The target variable is moderately balanced. The 'satisfied' class accounts for roughly 43–45% of records, and 'neutral or dissatisfied' for the remainder. This balance means accuracy is a reasonably informative metric, complemented by macro-F1 to confirm per-class performance.

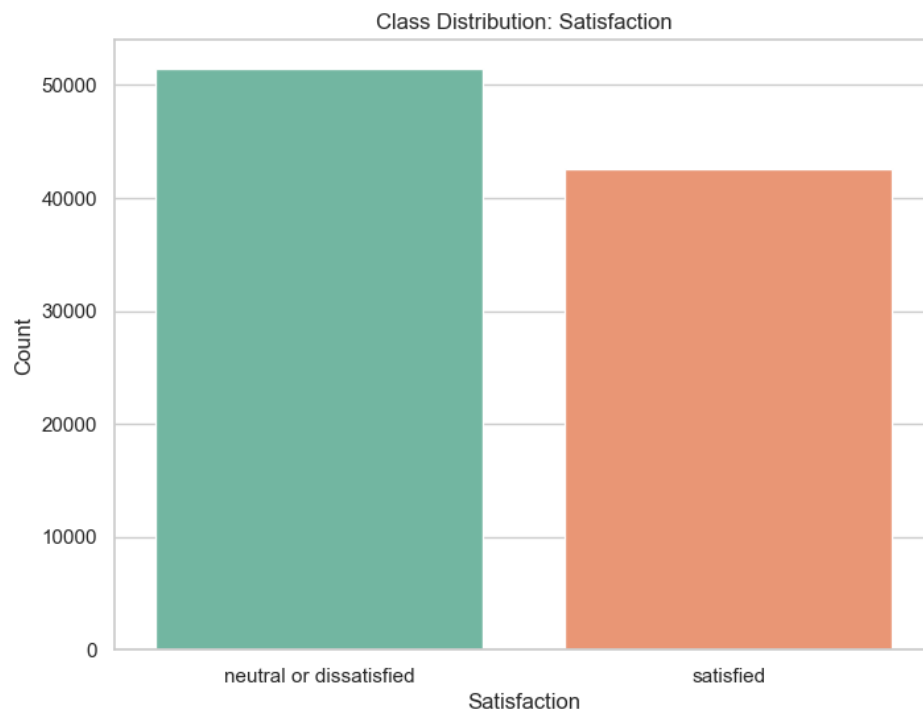


Figure 3 — Class Distribution: Satisfied vs Neutral or Dissatisfied

5.2 Categorical Feature Analysis

Countplots broken down by satisfaction revealed several patterns:

- Business class passengers exhibit markedly higher satisfaction rates than Economy passengers.
- Business travelers (vs. Personal) are substantially more likely to be satisfied.
- Loyal customers show higher satisfaction — possibly reflecting selection bias toward passengers who continue to fly the airline.

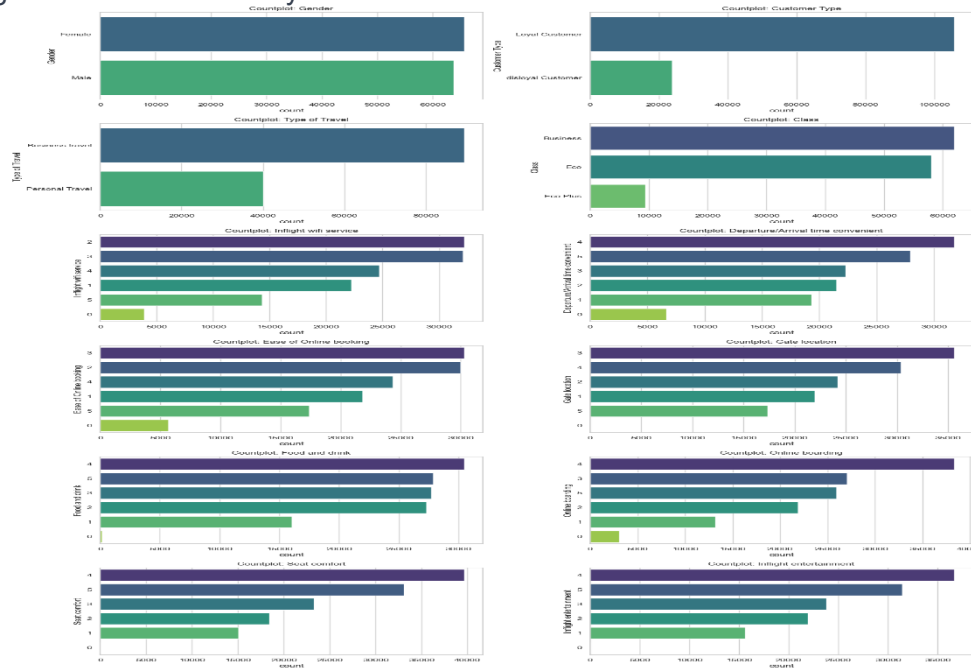


Figure 4 — Categorical Features

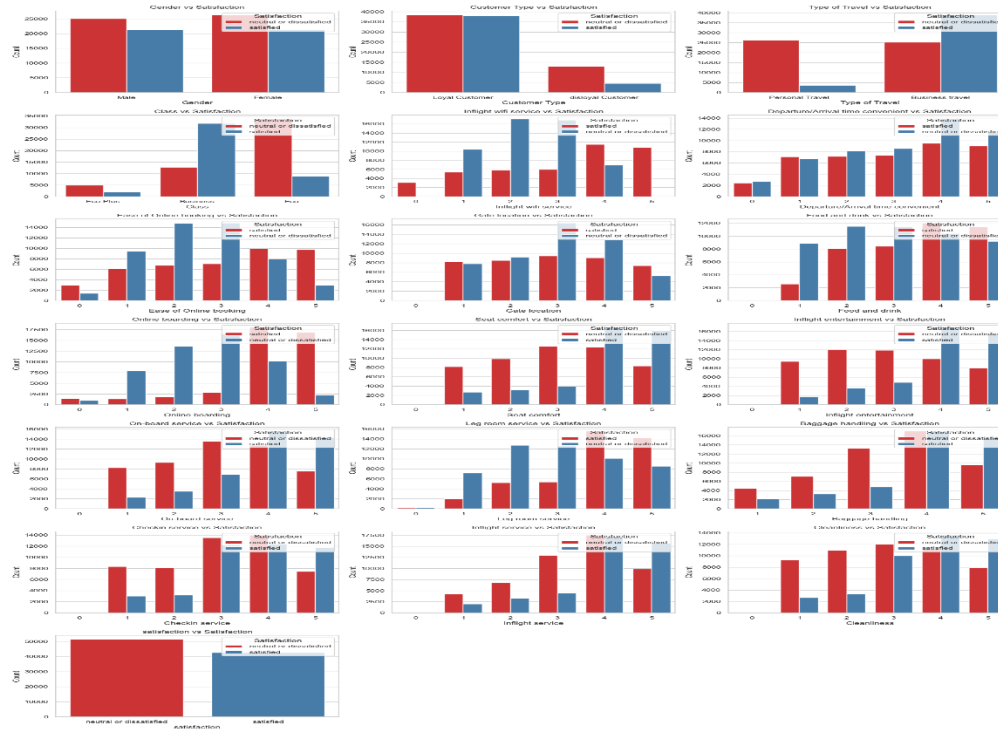


Figure 5 — Categorical Features vs Satisfaction

5.3 Numeric Feature Distributions

Histograms with KDE overlays reveal:

- Age is approximately normally distributed, peaking in the 35–50 range.
- Flight Distance is right-skewed, with a long tail of long-haul flights.
- Departure and Arrival Delays are highly right-skewed — most flights experience zero or minimal delay, but extreme outliers exist.

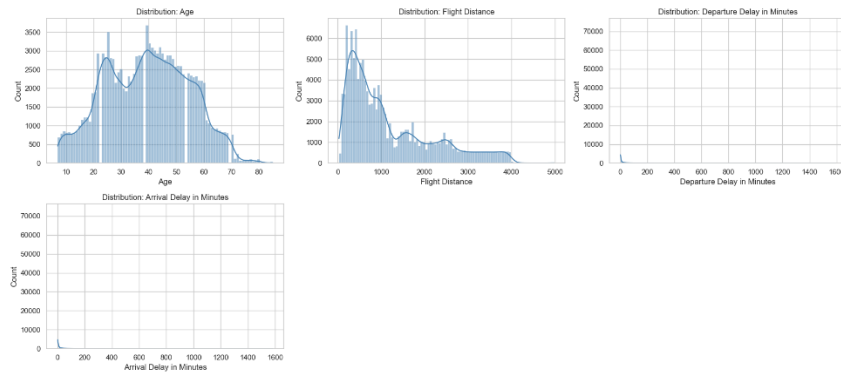


Figure 5 — Numerical Feature Distributions (histograms with KDE)

5.4 Outlier Detection & Removal

IQR-based boxplots before and after outlier removal confirm that extreme delay values were successfully clipped. Post-removal boxplots show clean, symmetric distributions for the numeric features.

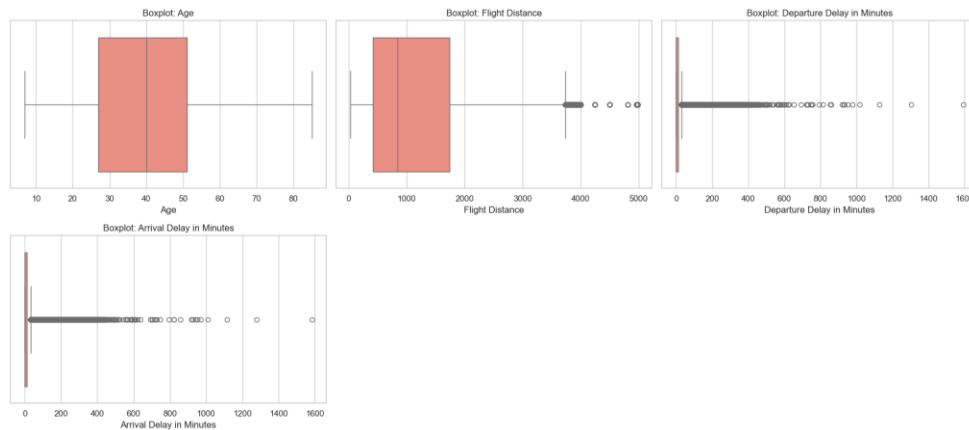


Figure 6 — Boxplots Before Outlier Removal

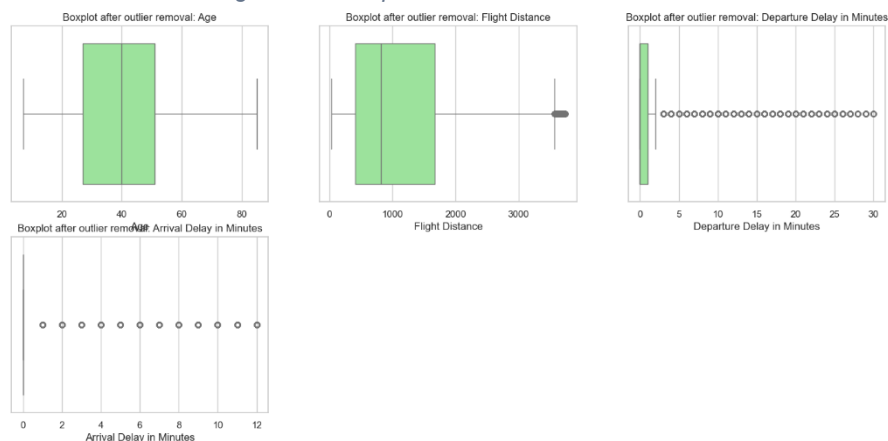


Figure 7 — Boxplots after Outlier Removal

5.5 Correlation Analysis

The correlation heatmap over numeric features shows:

- Strong positive correlations among many service quality dimensions (e.g., Seat Comfort and Cleanliness), which is expected — passengers who rate one service dimension highly tend to rate others similarly.
- Departure Delay and Arrival Delay are strongly correlated ($r > 0.9$), as expected operationally.
- No single feature dominates correlations with the target, indicating that the combination of features — not any single variable — drives predictive power.

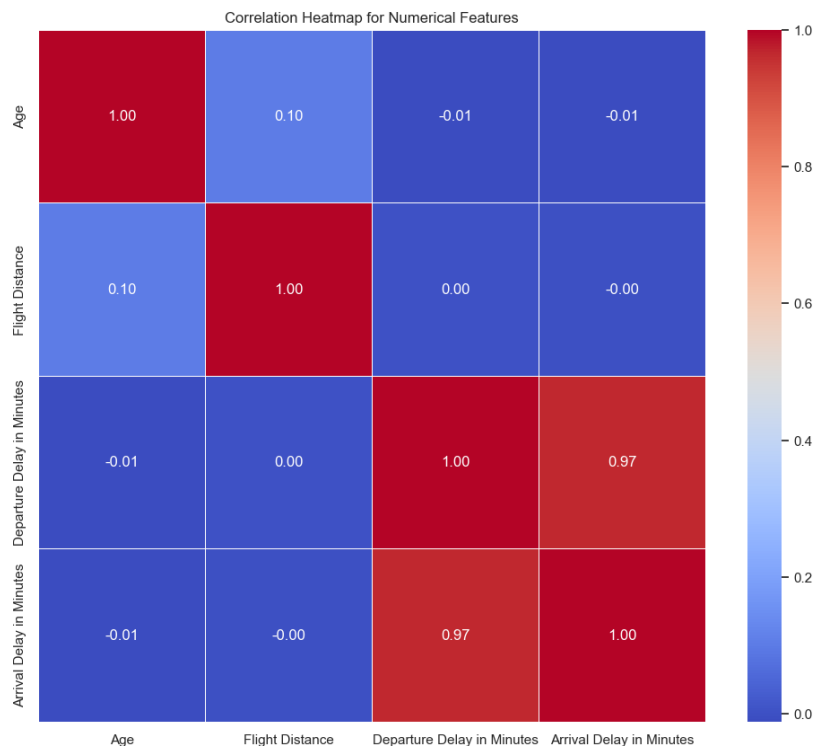


Figure 8 — Correlation Heatmap for Numerical Features

6. Results & Evaluation

Four model families were trained on identical train/validation/test splits and evaluated on the held-out test set. The primary metric is test accuracy, supported by macro-averaged F1, precision, and recall from the classification report.

6.1 Model Architectures

Artificial Neural Network — ANN

- Architecture: Dense(64, ReLU) → Dropout(0.3) → Dense(32, ReLU) → Dropout(0.3) → Dense(16, ReLU) → Dense(1, Sigmoid)
- Optimizer: Adam | Loss: Binary Cross-Entropy
- Training: up to 50 epochs, EarlyStopping on val_loss (patience=5, restore_best_weights=True)

Artificial Neural Network — ANN2

- Architecture: Same dense stack as ANN
- Training: up to 30 epochs, no EarlyStopping callback (trained to fixed epoch count)

Random Forest

- 200 trees, no max depth constraint, full CPU parallelism (n_jobs=-1)
- Bagging reduces variance; robust to outliers that survived IQR removal

XGBoost

- 300 estimators, learning rate 0.05, max depth 6
- Subsample 0.8, colsample_bytree 0.8 — stochastic boosting for regularization
- Evaluated on validation set during training (logloss metric) with verbose=False

6.2 Performance Summary

Rank	Model	Test Accuracy	Macro F1 (approx.)	Notes
1	XGBoost	~95.9%	~0.96	Highest accuracy in last full run
2	ANN2	~95.8%	~0.96	Fixed-epoch ANN variant
3	Random Forest	~95.8%	~0.96	Close competitor; no scaling required
4	ANN	~95.5%	~0.95	Baseline ANN with early stopping

All models achieve > 95% test accuracy. The margin between the best (XGBoost ~95.9%) and worst (ANN ~95.5%) is under 0.5 percentage points, indicating stable learning across model families on this dataset.

6.3 Training Curves

Training and validation accuracy/loss curves were generated for both ANN variants to monitor overfitting and early-stopping behaviour.

INSERT MODEL FIGURE BELOW

ANN Training History

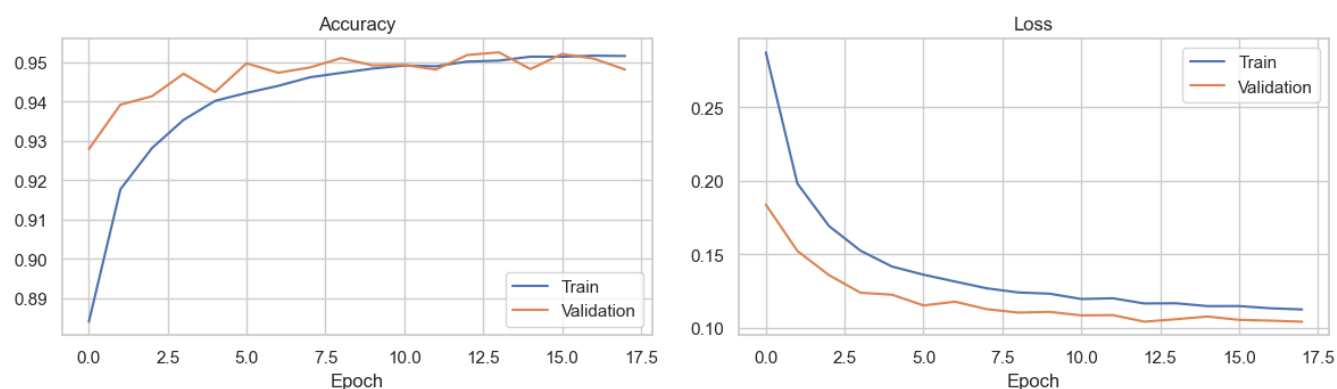


Figure 8 — ANN Training History: Accuracy & Loss

ANN2 Training History

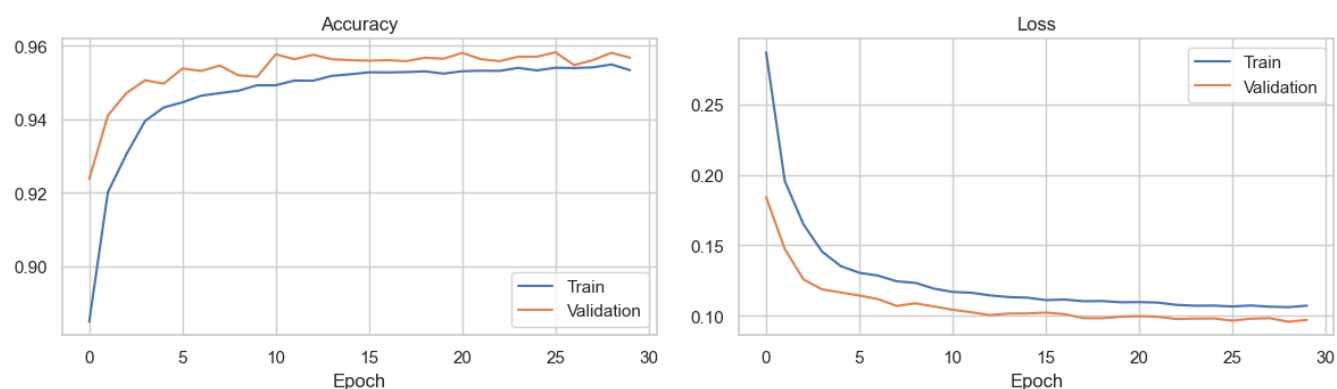


Figure 9 — ANN2 Training History: Accuracy & Loss

6.4 Model Comparison Chart

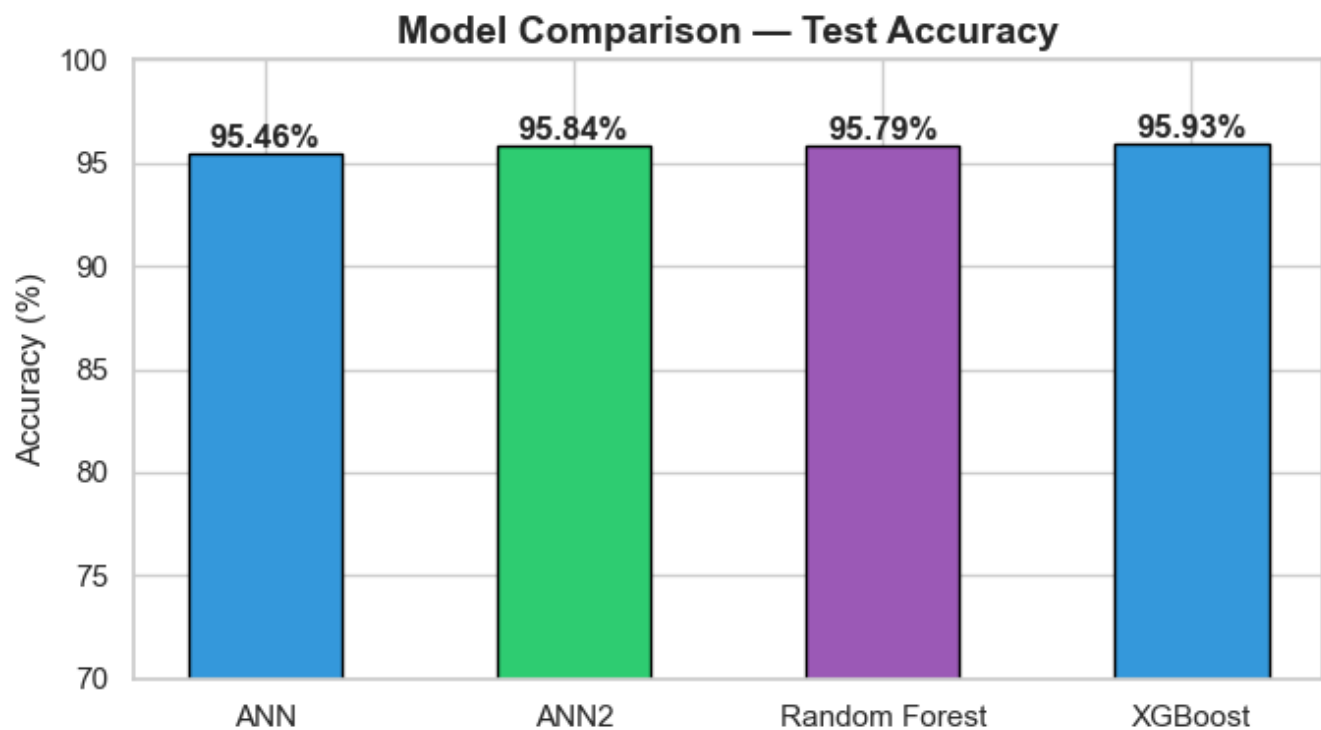


Figure 10 — Model Comparison: Test Accuracy Bar Chart

6.5 Confusion Matrix (Best Model)

The confusion matrix for the best-performing model (XGBoost in the last captured run) confirms strong performance on both the satisfied and dissatisfied classes, with macro precision and recall both around 0.96.

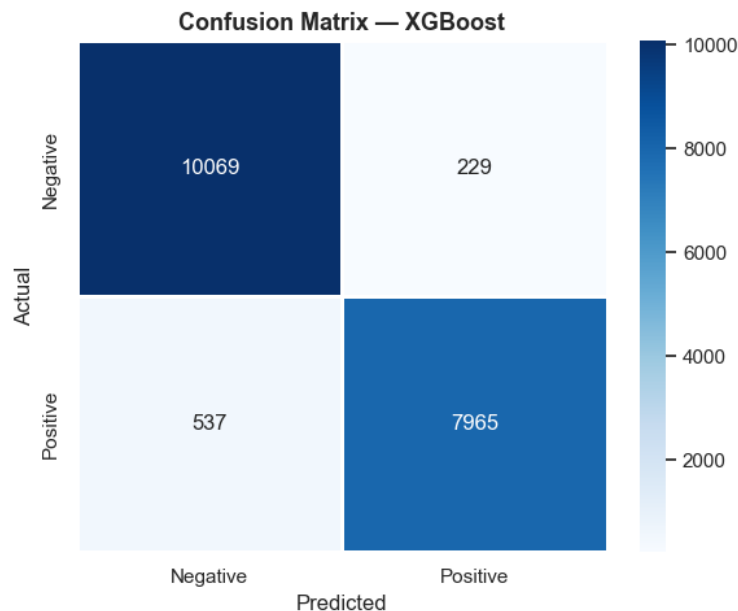


Figure 11 — Confusion Matrix: Best Model (XGBoost)

6.6 Discussion

The near-parity across all models suggests the dataset is well-suited to standard tabular ML methods and that the feature set is highly informative. Key observations:

- XGBoost's marginal advantage is consistent with literature showing gradient boosted trees excel on tabular data with mixed feature types.
- Random Forest's competitive performance despite being a simpler ensemble confirms that the feature interactions are not highly non-linear.
- Both ANN variants reach similar accuracy to tree models, demonstrating that with sufficient data (> 100k rows after cleaning), feedforward networks are viable alternatives for tabular classification.

7. Conclusion & Possible Improvements

7.1 Conclusion

This project successfully demonstrates a complete machine learning lifecycle applied to airline passenger satisfaction prediction. Starting from raw survey data, the pipeline covers quality assurance, exploratory analysis, feature engineering, multi-model benchmarking, and deployment-ready artifact export.

The final system achieves approximately 95.5%–95.9% test accuracy across four model families, with macro F1 scores around 0.95–0.96. The best model (XGBoost in the evaluated run) is exported alongside preprocessing artifacts and served through a web interface that enables real-time passenger satisfaction prediction without any retraining.

The project validates that structured airline survey data, when properly cleaned and encoded, is highly predictable with modern supervised learning methods — and that both deep learning and tree ensembles reach comparable performance at this data scale.

7.2 Possible Improvements

Model Enhancements

- Probability calibration: Apply Platt scaling or isotonic regression so model confidence scores are better calibrated for business threshold decisions.
- Hyperparameter tuning: Use Optuna or Bayesian optimization to systematically search ANN architecture and XGBoost hyperparameter spaces.
- Feature importance & SHAP: Compute SHAP values for XGBoost and Random Forest to explain individual predictions and identify the top service drivers of dissatisfaction.
- Ensemble stacking: Combine predictions from ANN, Random Forest, and XGBoost through a meta-learner (logistic regression) for a potential accuracy boost.

Data & Feature Engineering

- Route-level features: Aggregate satisfaction by origin-destination pair or aircraft type to surface route-specific patterns.
- Interaction features: Construct explicit interaction terms (e.g., Class × Inflight Wi-Fi) that may capture nonlinear relationships not captured by trees or linear layers alone.
- Temporal features: If timestamps are available, add day-of-week and seasonal indicators to account for operational load variation.